

Scaling Book — Section 7 Exercises

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Pop Quiz

Take gen step on BS of 4 tokens on 30B param dense model on TPU v5e 4×4 slice in int8 w/ bf16 FLOPs, 8192 context, 100 kB/token KV caches.

(i)

$$T_{\min, \text{step}} = \frac{4 \cdot 100 \cdot 10^3 \cdot 8192}{4 \cdot 4 \cdot 8.2 \cdot 10^{11}} + \max \left(\underbrace{\frac{2 \cdot 4 \cdot 30 \cdot 10^9}{4 \cdot 4 \cdot 1.97 \cdot 10^{14}}}_{7.6 \cdot 10^{-5} \text{ s}}, \underbrace{\frac{30 \cdot 10^9}{4 \cdot 4 \cdot 8.2 \cdot 10^{11}}}_{2.2 \text{ ms}} \right) \\ \approx 2.5 \text{ ms.}$$

MLPs are comm-bounded.

We could also see this since our total token BS is 4 per fwd pass $\ll B_{\text{crit}} = 240$.

(ii)

$B = 256 > B_{\text{crit}} = 240 \Rightarrow$ MLPs are comp bound.

For thoroughness:

$$T_{\min, \text{step}} = \underbrace{\frac{256 \cdot 100 \cdot 10^3 \cdot 8096}{16 \cdot 8.2 \cdot 10^{11}}}_{\approx 16 \text{ ms}} + \max \left(\underbrace{\frac{2 \cdot 256 \cdot 30 \cdot 10^9}{16 \cdot 1.97 \cdot 10^{14}}}_{\approx 4.9 \text{ ms}}, \underbrace{\frac{30 \cdot 10^9}{16 \cdot 8.2 \cdot 10^{11}}}_{\approx 4.6 \text{ ms}} \right) \\ = 20.9 \text{ ms.}$$

Q1

(i)

Per layer:

$$\begin{aligned} \text{Attn: } & 2DNH + 2DKH + D \approx 2DNH + 2DKH \\ \text{MLP: } & 3DF \end{aligned}$$

$$\begin{aligned}
\text{Total} &= L(2DNH + 2DKH + 3DF) + DV \\
&= 64 \cdot (2 \cdot 4096 \cdot 32 \cdot 256 + 2 \cdot 4096 \cdot 8 \cdot 256 + 3 \cdot 4096 \cdot 16384) \\
&\quad + 4096 \cdot 32128 \approx 18.4\text{B params.}
\end{aligned}$$

(ii)

$$\text{KV cache size/token} = 2 \cdot H \cdot K \cdot L = 2 \cdot 256 \cdot 8 \cdot 64 = 262,144 \text{ bytes} \approx 262 \text{ kB.}$$

Q2

Serve on TPU v5e 4×4 slice. int8, $T = 128\text{K}$.

(i)

$$\begin{aligned}
\text{KV Cache Size/seq} &= 2HKL T = 2 \cdot 256 \cdot 8 \cdot 64 \cdot 128 \cdot 10^3 \\
&\approx 3.4 \cdot 10^{10} \text{ bytes} = 34 \text{ GB.}
\end{aligned}$$

$$\text{Model size} = 18.4 \cdot 10^9 \text{ bytes} = 18.4 \text{ GB.}$$

$$\text{Total mem} = 16 \cdot 16 \text{ GB} = 256 \text{ GB.}$$

$$\text{BS}_{\max} = \frac{256 - 18.4}{34} \approx 7.$$

(ii)

Dropping $K = 1$ scales down KV cache size/token by $\frac{1}{8} \Rightarrow \text{BS}_{\max} = 8 \cdot 7 = 56$.

Q3

$$\frac{\text{Model Size}}{(\# \text{ devices}) \cdot W_{\text{hbm}}} = \frac{18.4 \cdot 10^9}{16 \cdot 8.2 \cdot 10^{11}} \approx 1 \text{ ms} \quad (\text{assuming full utilization of HBM}).$$

Q4

What does ICI look like on a 4×4 ? $\Rightarrow$ 2 hardware axes, doubled bw.

Roofline bound on TP:

$$\frac{2 \cdot F}{\left(\frac{C}{W_{\text{ici}}}\right)} = \frac{2 \cdot 16384 \cdot 9.0 \cdot 10^{10}}{3.94 \cdot 10^{14}} = 7 < Y = 16.$$

We are comm-bounded with TP only.

Prefill: From our TP roofline, if we use the whole slice for TP, we will be comm-bounded. What I would do is use 8-way TP which will still be slightly comm-bounded but right on the edge, with 2-way sequence parallelism.

Generation: For generation, our only option is model parallelism. Per layer, since a majority of the parameters are within the MLP (approximating by ignoring einsum):

$$T_{\text{hbm comm}} = \frac{2DF}{16 \cdot W_{\text{hbm}}}, \quad T_{\text{ici comm}} = \frac{2BD}{W_{\text{ici}}}.$$

Want:

$$\begin{aligned} T_{\text{ici comm}} < T_{\text{hbm comm}} &\Rightarrow \frac{2BD}{W_{\text{ici}}} < \frac{2DF}{16 \cdot W_{\text{hbm}}} \\ B < \frac{F \cdot W_{\text{ici}}}{16 \cdot W_{\text{hbm}}} &= \frac{16384 \cdot 9.0 \cdot 10^{10}}{16 \cdot 8.2 \cdot 10^{11}} \\ &\Rightarrow B < 112 \end{aligned}$$

Which will probably be true in practice since 112 is a pretty big batch size of requests to arrive at once.

For generation, I'd use 16-way TP since we're already comm-bounded and as shown above at reasonable inference batch sizes our ici comm time will be less than the time to load from HBM $\rightarrow$ MXU. Since K (# of KV heads) is < 16 but divides it, I'd shard the KV cache along the head and batch dimensions.

Under this sharding each chip would store $\frac{1}{16}$ of all the parameters so assuming BS = 1:

$$\text{Step Time}_{\min} = \frac{(34 + 18.4) \cdot 10^9}{16 \cdot 8.2 \cdot 10^{11}} \approx 3 \text{ ms}.$$

This comes from the fact that we're still bottlenecked by loading params from HBM and each device has roughly (1/16) of the data.

Q5

$E = 16, k = 2$.

(i)

Per layer:

$$\begin{aligned} \text{Attn} &= 2DNH + 2DKH \\ \text{MLP}_{\text{total}} &= 3DF \cdot E \\ \text{MLP}_{\text{act}} &= 3DF \cdot k \end{aligned}$$

$$\begin{aligned} \text{Total} &= 2DV + L \cdot (2DNH + 2DKH + 3DFE) \\ &= 2 \cdot 4096 \cdot 32128 + 64 \cdot (2 \cdot 4096 \cdot 32 \cdot 256 \\ &\quad + 2 \cdot 4096 \cdot 8 \cdot 256 + 3 \cdot 4096 \cdot 16384 \cdot 16) \\ &\approx 211\text{B}. \end{aligned}$$

$$\begin{aligned}\text{Activated} &= 2DV + L \cdot (2DNH + 2DKH + 3DFk) \\ &\approx 31B.\end{aligned}$$

(ii)

Since MLP accounts for most of the FLOPs, we see the bs on a single device where we become comm bound for one of the matmuls (all 3 are the same size). Also assuming we're in generation and our sequence dim is trivial.

$$\text{AI} = \frac{2BDFk}{2BD + 2DFE + 2BFk} \approx \frac{2BDFk}{2DFE} = B \cdot \frac{k}{E}.$$

$$B \cdot \frac{k}{E} \geq \text{Intensity(v5e)} = 240$$

$$\Rightarrow B \geq \frac{240 \cdot E}{k} = 1920.$$

(iii)

The KV Cache size per token stays the same as 262 kB since MoE doesn't change the K, V projections.

(iv)

Again, our linear projections dominate FLOPs so:

$$\text{FLOPS} \approx 2 \cdot (\text{activated params}) \cdot T = 2 \cdot 31 \cdot 10^9 \cdot T = 62 \cdot 10^9 \cdot T.$$

Q6

Each FFW weight sharded as $[E_z, D_x, F_y]$ (z only used during training as FSDP dim).

(i)

TPU v5e 8×16 slice, $Y = 8$, $Z = 16$, int 8.

$$T_{\text{weights}} \approx \frac{211 \cdot 10^9}{8 \cdot 16 \cdot W_{\text{hbm}}} \approx 2 \text{ ms} \quad (\text{most parameters are in the MLPs so this is a good approximation}).$$

$$\text{Per chip mem used} = \left(\frac{211 \cdot 10^9}{8 \cdot 16} \right) \approx 1.65 \text{ GB}.$$

$$\text{Remaining per chip mem} = 16 - 1.65 = 14.35 \text{ GB}.$$

(ii)

Again, assuming weights are stored in int8:

$$N_{\text{min}} = \frac{211}{16} = 14 \text{ v5es} \Rightarrow 7 \times 2 \text{ slice} \quad (\text{no wraparound connections though}).$$

Q7

For 2D sharding:

$$T_{\text{math}} = \left(\frac{2BDF}{XYZ} + \frac{2BDF}{XYZ} \right) / C = \frac{4BDF}{CXYZ}.$$

Assuming bf16, bidirectional ici:

$$\begin{aligned} T_{\text{comm}} &= \frac{2B \cdot \left(\frac{D}{X}\right)}{2 \cdot W_{\text{ici}}} + \frac{2 \cdot 2B \cdot \left(\frac{F}{YZ}\right)}{W_{\text{ici}}} + \frac{2B \cdot \left(\frac{D}{X}\right)}{2 \cdot W_{\text{ici}}} \\ &= \frac{4BD}{2XW_{\text{ici}}} + \frac{4BF}{YZW_{\text{ici}}}. \end{aligned}$$

Since we are free to set the topology, we want to minimize X, YZ (only the product matters for T_{comm}):

$$\begin{aligned} YZ &= N/X, \quad X = N/YZ. \\ T_{\text{comm}} &= \frac{4BD}{2XW_{\text{ici}}} + \frac{4BFX}{NW_{\text{ici}}}. \\ \frac{\partial T_{\text{comm}}}{\partial X} &= \frac{-4BD}{2W_{\text{ici}}X^2} + \frac{4BF}{NW_{\text{ici}}} = 0 \\ \Rightarrow \frac{4BF}{NW_{\text{ici}}} &= \frac{4BD}{2W_{\text{ici}}X^2} \Rightarrow X^2 = \frac{D}{2W_{\text{ici}}} \cdot \frac{NW_{\text{ici}}}{F} = \frac{ND}{2F}. \\ X_{\text{opt}} &= \sqrt{\frac{ND}{2F}}, \quad YZ_{\text{opt}} = \frac{N}{X} = \sqrt{\frac{N^2 \cdot 2F}{ND}} = \sqrt{\frac{2FN}{D}}. \end{aligned}$$

Assuming $F = 4D$ like in specified model:

$$X_{\text{opt}} = \sqrt{\frac{ND}{2 \cdot 4D}} = \sqrt{\frac{N}{8}}.$$

$$\begin{aligned} T_{\text{comm}} &= \frac{4BD}{2 \cdot \sqrt{N/8} \cdot W_{\text{ici}}} + \frac{4BF \cdot \sqrt{N/8}}{N \cdot W_{\text{ici}}} \\ &= \frac{2BD}{W_{\text{ici}} \cdot \sqrt{N/8}} + \frac{16BD \cdot \sqrt{N/8}}{N \cdot W_{\text{ici}}} \\ &= \frac{2BD}{W_{\text{ici}}} \cdot \left(\sqrt{\frac{8}{N}} + 8 \cdot \sqrt{\frac{N}{8N^2}} \right) \\ &= \frac{2BD}{W_{\text{ici}}} \left(\sqrt{\frac{8}{N}} + \sqrt{\frac{64N}{8N^2}} \right) = \frac{4BD}{W_{\text{ici}}} \cdot \sqrt{\frac{8}{N}} \\ &= \frac{\sqrt{128} \cdot BD}{W_{\text{ici}} \cdot \sqrt{N}}. \end{aligned}$$

For normal 3D model parallelism ($[D, F_{XYZ}]$):

$$T_{\text{math}} = \frac{2BDF}{XYZ \cdot C} \cdot 2 = \frac{4BDF}{XYZ \cdot C}.$$

$$T_{\text{comm}} = \frac{2 \cdot 2 \cdot BD}{3 \cdot W_{\text{ici}}} = \frac{4BD}{3W_{\text{ici}}}.$$

The time for math comp is the same so we're looking for when:

$$\begin{aligned} T_{\text{comm}}^{3D} > T_{\text{comm}}^{2D} &\Rightarrow \frac{4BD}{3W_{\text{ici}}} > \frac{\sqrt{128} \cdot BD}{W_{\text{ici}} \cdot \sqrt{N}} \\ \Rightarrow \sqrt{N} > \frac{3 \cdot \sqrt{128}}{4} &\Rightarrow N > \frac{9 \cdot 128}{16} = 72. \end{aligned}$$

When we have more than 72 chips in our slice (assuming slice is chosen s.t. we have bidirectional ici over all axes), the 2D scheme will outperform the 3D one.